

1NF (First Normal Form) – Remove multi-valued attributes:

StudentID	StudentName	Address	CourseCode	CourseName	InstructorName
1001	John Smith	123 Main St	CSE101	Intro to CS	Dr. Adams
1001	John Smith	123 Main St	MTH102	Calculus I	Dr. Clark
1002	Jane Doe	456 Maple Ave	PHY103	Physics I	Dr. Thomas
1003	Alice Brown	789 Oak Lane	CSE101	Intro to CS	Dr. Adams
1003	Alice Brown	789 Oak Lane	PHY103	Physics I	Dr. Thomas

2NF (Second Normal Form) – Remove partial dependencies:

Split to;

Student Table:

StudentID	StudentName	Address
1001	John Smith	123 Main St
1002	Jane Doe	456 Maple Ave
1003	Alice Brown	789 Oak Lane

Course Table:

CourseCode	CourseName	InstructorName
CSE101	Intro to CS	Dr. Adams
MTH102	Calculus I	Dr. Clark
PHY103	Physics I	Dr. Thomas

Enrollment Table:

StudentID	CourseCode
1001	CSE101
1001	MTH102
1002	PHY103
1003	CSE101
1003	PHY103

3NF (Third Normal Form) – Remove transitive dependencies:

All tables above are already in 3NF:

- No transitive dependencies.
- Each non-key depends only on the primary key.